

Learning Activity Sheet for General Science

Quarter 3

Lesson

3

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Learning Activity Sheet for Science Grade 11
Quarter 3: Lesson 3

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LEARNING ACTIVITY SHEET

Learning Area:	General Science	Quarter:	3
Lesson Exemplar No.:	3	Date:	
Lesson Title/ Topic:	The Importance of Cells in Living Things		
Name:		Grade & Section:	

I. Activity No. 1: Observing Cellular Respiration in Action

II. Objective(s): At the end of the activity, you should be able to:

- Describe how cells obtain nutrients and convert them into energy.
- Demonstrate the process of cellular respiration using a simple yeast experiment.
- Relate the results of the activity to how living organisms sustain life.

III. Materials Needed: 1 small plastic bottle or test tube

- 1 packet (1 tablespoon) of dry yeast
- 2 tablespoons of sugar
- $\frac{1}{2}$ cup of warm water (not hot)
- 1 balloon
- 1 funnel (optional)
- Spoon and measuring cup
- Marker or pen

IV. Instructions:

- Label your bottle "Yeast Respiration Setup."
- Pour $\frac{1}{2}$ cup of warm water into the bottle.
- Add 2 tablespoons of sugar and stir until dissolved.
- Add 1 tablespoon of yeast using a funnel. Swirl gently to mix.
- Quickly stretch the balloon over the mouth of the bottle.
- Place the setup in a warm area and observe for 15–20 minutes.
- Record your observations using the table below.

Observation Table

Time (minutes)	Observation (Balloon Size / Bubbles Formed)	Inference
0		
5		

10		
15		
20		

Safety Reminders

- Use only warm, not hot, water to avoid killing the yeast.
- Do not taste or consume the mixture.
- Wash your hands after the experiment and dispose of the mixture properly.

RUBRICS for Observation and Inference			
3	2	1	0
Demonstrates clear and accurate understanding of the topic; expresses one big idea that captures the essence of the lesson.	Shows partial understanding ; expresses a relevant but incomplete or somewhat vague idea.	Shows limited understanding ; idea is unclear or contains misconceptions.	No response or response unrelated to the topic.

V. Synthesis:

1. What did you observe in the yeast setup after several minutes?

2. What gas caused the balloon to inflate?

3. How does this activity show that living cells use nutrients to release energy?

4. In what ways is yeast similar to or different from human and plant cells in obtaining energy?

5. Why is cellular respiration important for the survival of living things?
